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API Rate Limiter Simulation

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1. Problem Statement

Web applications often rely on APIs to fetch or send data. If users trigger actions repeatedly (such as rapid button clicks), the system may generate too many API requests in a short time.

This can cause:

- Server overload
- Performance degradation
- Unnecessary resource consumption

Therefore, a mechanism is needed to **limit the number of function executions within a specific time interval.**



2. Solution

The solution is an **API Rate Limiter Simulation** implemented using JavaScript.

Key features of the system:

- A **throttle function** that limits function execution per second
- A **user-defined rate limit** (calls per second)
- Real-time display of **allowed and blocked calls**
- Logging of execution timestamps

This system demonstrates how throttling can control execution frequency and simulate real-world API rate limiting.



3. Working (Method / Mind Map)

Set Rate Limit

User sets the number of allowed API calls per second.

Throttle Logic

Throttle function checks call count using `Date.now()` to control execution frequency.

Allow / Block Calls

Calls within limit are allowed. Excess calls are blocked and logged.



4. Conclusion

This project demonstrates how **throttling can be used to implement API rate limiting**.

Key outcomes:

- Controlled function execution frequency
- Prevention of excessive API calls
- Real-time visualization of allowed and blocked requests

The system shows how timing logic, closures, and higher-order functions can be combined to build efficient request control mechanisms in web applications





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Thank You